

Cables and Arches

Cables and arches often form the main load-carrying element in many types of structures, and in this chapter we will discuss some of the important aspects related to their structural analysis. The chapter begins with a general discussion of cables, followed by an analysis of cables subjected to a concentrated load and to a uniform distributed load. Since most arches are statically indeterminate, only the special case of a three-hinged arch will be considered. The analysis of this structure will provide some insight regarding the fundamental behavior of all arched structures.

5.1 Cables

Cables are often used in engineering structures for support and to transmit loads from one member to another. When used to support suspension roofs, bridges, and trolley wheels, cables form the main load-carrying element in the structure. In the force analysis of such systems, the weight of the cable itself may be neglected; however, when cables are used as guys for radio antennas, electrical transmission lines, and derricks, the cable weight may become important and must be included in the structural analysis. Two cases will be considered in the sections that follow: a cable subjected to concentrated loads and a cable subjected to a distributed load. Provided these loadings are coplanar with the cable, the requirements for equilibrium are formulated in an identical manner.

When deriving the necessary relations between the force in the cable and its slope, we will make the assumption that the cable is *perfectly flexible* and *inextensible*. Due to its flexibility, the cable offers no resistance to shear or bending and, therefore, the force acting in the cable is always tangent to the cable at points along its length. Being inextensible, the cable has a constant length both before and after the load is applied. As a result, once the load is applied, the geometry of the cable remains fixed, and the cable or a segment of it can be treated as a rigid body.

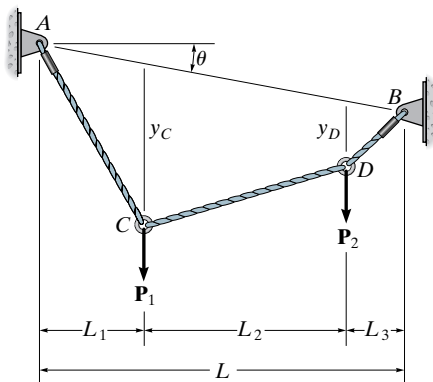


Fig. 5-1



The deck of this harp cable-stayed bridge is supported by a series of cables attached at various points along the deck and pylons.

5.2 Cable Subjected to Concentrated Loads

When a cable of negligible weight supports several concentrated loads, the cable takes the form of several straight-line segments, each of which is subjected to a constant tensile force. Consider, for example, the cable shown in Fig. 5-1. Here θ specifies the angle of the cable's cord AB , and L is the cable's span. If the distances L_1 , L_2 , and L_3 and the loads $\mathbf{P}_1$ and $\mathbf{P}_2$ are known, then the problem is to determine the *nine unknowns* consisting of the tension in each of the three segments, the four components of reaction at A and B , and the sags y_C and y_D at the two points C and D . For the solution we can write *two* equations of force equilibrium at each of points A , B , C , and D . This results in a total of *eight equations*. To complete the solution, it will be necessary to know something about the geometry of the cable in order to obtain the necessary ninth equation. For example, if the cable's total length $\mathcal{L}$ is specified, then the Pythagorean theorem can be used to relate $\mathcal{L}$ to each of the three segmental lengths, written in terms of θ , y_C , y_D , L_1 , L_2 , and L_3 . Unfortunately, this type of problem cannot be solved easily by hand. Another possibility, however, is to specify one of the sags, either y_C or y_D , instead of the cable length. By doing this, the equilibrium equations are then sufficient for obtaining the unknown forces and the remaining sag. Once the sag at each point of loading is obtained, $\mathcal{L}$ can then be determined by trigonometry.

When performing an equilibrium analysis for a problem of this type, the forces in the cable can also be obtained by writing the equations of equilibrium for the entire cable or any portion thereof. The following example numerically illustrates these concepts.

EXAMPLE 5.1

Determine the tension in each segment of the cable shown in Fig. 5-2a. Also, what is the dimension h ?

SOLUTION

By inspection, there are four unknown external reactions (A_x , A_y , D_x , and D_y) and three unknown cable tensions, one in each cable segment. These seven unknowns along with the sag h can be determined from the eight available equilibrium equations ($\sum F_x = 0$, $\sum F_y = 0$) applied to points A through D .

A more direct approach to the solution is to recognize that the slope of cable CD is specified, and so a free-body diagram of the entire cable is shown in Fig. 5-2b. We can obtain the tension in segment CD as follows:

$$\zeta + \sum M_A = 0;$$

$$T_{CD}(3/5)(2 \text{ m}) + T_{CD}(4/5)(5.5 \text{ m}) - 3 \text{ kN}(2 \text{ m}) - 8 \text{ kN}(4 \text{ m}) = 0$$

$$T_{CD} = 6.79 \text{ kN} \quad \text{Ans.}$$

Now we can analyze the equilibrium of points C and B in sequence. Point C (Fig. 5-2c);

$$\rightarrow \sum F_x = 0; \quad 6.79 \text{ kN} (3/5) - T_{BC} \cos \theta_{BC} = 0$$

$$+\uparrow \sum F_y = 0; \quad 6.79 \text{ kN} (4/5) - 8 \text{ kN} + T_{BC} \sin \theta_{BC} = 0$$

$$\theta_{BC} = 32.3^\circ \quad T_{BC} = 4.82 \text{ kN} \quad \text{Ans.}$$

Point B (Fig. 5-2d);

$$\rightarrow \sum F_x = 0; \quad -T_{BA} \cos \theta_{BA} + 4.82 \text{ kN} \cos 32.3^\circ = 0$$

$$+\uparrow \sum F_y = 0; \quad T_{BA} \sin \theta_{BA} - 4.82 \text{ kN} \sin 32.3^\circ - 3 \text{ kN} = 0$$

$$\theta_{BA} = 53.8^\circ \quad T_{BA} = 6.90 \text{ kN} \quad \text{Ans.}$$

Hence, from Fig. 5-2a,

$$h = (2 \text{ m}) \tan 53.8^\circ = 2.74 \text{ m} \quad \text{Ans.}$$

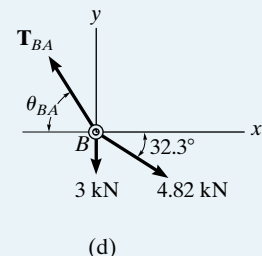
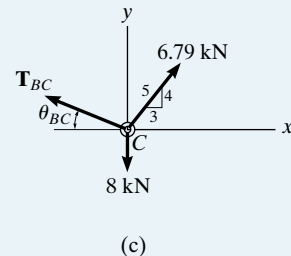
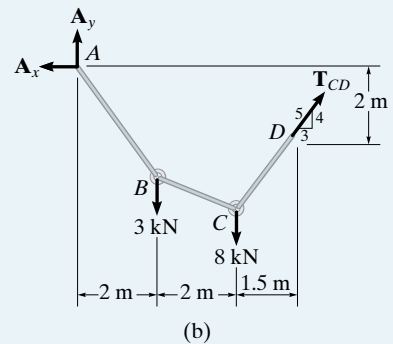
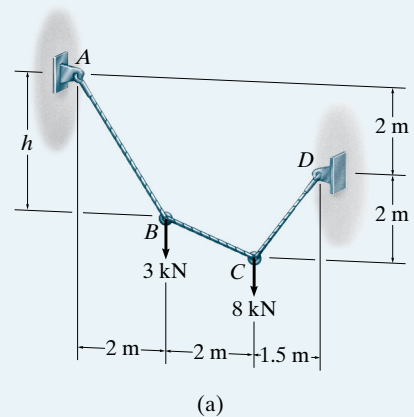


Fig. 5-2

5.3 Cable Subjected to a Uniform Distributed Load

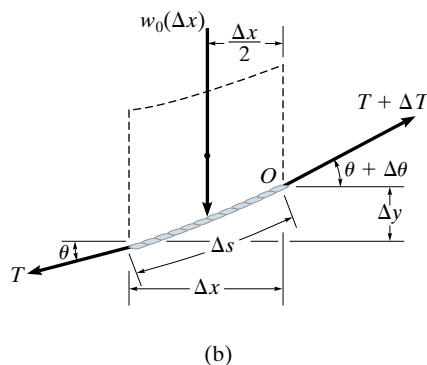
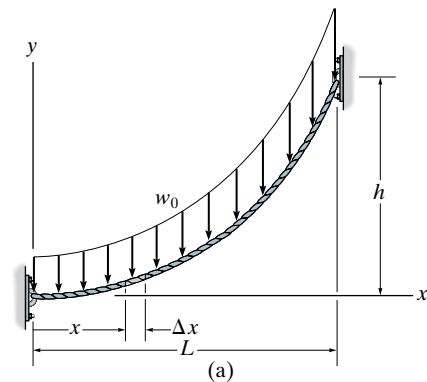


Fig. 5-3

Cables provide a very effective means of supporting the dead weight of girders or bridge decks having very long spans. A suspension bridge is a typical example, in which the deck is suspended from the cable using a series of close and equally spaced hangers.

In order to analyze this problem, we will first determine the shape of a cable subjected to a uniform *horizontally* distributed vertical load w_0 , Fig. 5-3a. Here the x, y axes have their origin located at the lowest point on the cable, such that the slope is zero at this point. The free-body diagram of a small segment of the cable having a length Δs is shown in Fig. 5-3b. Since the tensile force in the cable changes continuously in both magnitude and direction along the cable's length, this change is denoted on the free-body diagram by ΔT . The distributed load is represented by its resultant force $w_0\Delta x$, which acts at $\Delta x/2$ from point O . Applying the equations of equilibrium yields

$$\rightarrow \Sigma F_x = 0; \quad -T \cos \theta + (T + \Delta T) \cos(\theta + \Delta\theta) = 0$$

$$+\uparrow \Sigma F_y = 0; \quad -T \sin \theta - w_0(\Delta x) + (T + \Delta T) \sin(\theta + \Delta\theta) = 0$$

$$\zeta + \Sigma M_O = 0; \quad w_0(\Delta x)(\Delta x/2) - T \cos \theta \Delta y + T \sin \theta \Delta x = 0$$

Dividing each of these equations by Δx and taking the limit as $\Delta x \rightarrow 0$, and hence $\Delta y \rightarrow 0$, $\Delta\theta \rightarrow 0$, and $\Delta T \rightarrow 0$, we obtain

$$\frac{d(T \cos \theta)}{dx} = 0 \quad (5-1)$$

$$\frac{d(T \sin \theta)}{dx} = w_0 \quad (5-2)$$

$$\frac{dy}{dx} = \tan \theta \quad (5-3)$$

Integrating Eq. 5-1, where $T = F_H$ at $x = 0$, we have:

$$T \cos \theta = F_H \quad (5-4)$$

which indicates the horizontal component of force at *any point* along the cable remains *constant*.

Integrating Eq. 5-2, realizing that $T \sin \theta = 0$ at $x = 0$, gives

$$T \sin \theta = w_0 x \quad (5-5)$$

Dividing Eq. 5-5 by Eq. 5-4 eliminates T . Then using Eq. 5-3, we can obtain the slope at any point,

$$\tan \theta = \frac{dy}{dx} = \frac{w_0 x}{F_H} \quad (5-6)$$

Performing a second integration with $y = 0$ at $x = 0$ yields

$$y = \frac{w_0}{2F_H} x^2 \quad (5-7)$$

This is the equation of a *parabola*. The constant F_H may be obtained by using the boundary condition $y = h$ at $x = L$. Thus,

$$F_H = \frac{w_0 L^2}{2h} \quad (5-8)$$

Finally, substituting into Eq. 5-7 yields

$$y = \frac{h}{L^2} x^2 \quad (5-9)$$

From Eq. 5-4, the maximum tension in the cable occurs when θ is maximum; i.e., at $x = L$. Hence, from Eqs. 5-4 and 5-5,

$$T_{\max} = \sqrt{F_H^2 + (w_0 L)^2} \quad (5-10)$$

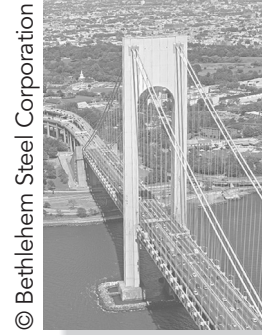
Or, using Eq. 5-8, we can express $T_{\max}$ in terms of w_0 , i.e.,

$$T_{\max} = w_0 L \sqrt{1 + (L/2h)^2} \quad (5-11)$$

Realize that we have neglected the weight of the cable, which is *uniform* along the *length* of the cable, and not along its horizontal projection. Actually, a cable subjected to its own weight and free of any other loads will take the form of a *catenary curve*. However, if the sag-to-span ratio is small, which is the case for most structural applications, this curve closely approximates a parabolic shape, as determined here.

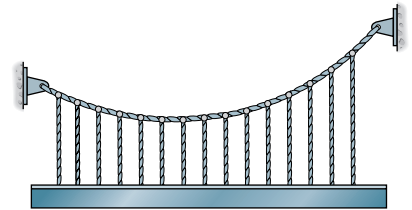
From the results of this analysis, it follows that a cable will *maintain a parabolic shape*, provided the dead load of the deck for a suspension bridge or a suspended girder will be *uniformly distributed* over the horizontal projected length of the cable. Hence, if the girder in Fig. 5-4a is supported by a series of hangers, which are close and uniformly spaced, the load in each hanger must be the *same* so as to ensure that the cable has a parabolic shape.

Using this assumption, we can perform the structural analysis of the girder or any other framework which is freely suspended from the cable. In particular, if the girder is simply supported as well as supported by the cable, the analysis will be statically indeterminate to the first degree, Fig. 5-4b. However, if the girder has an internal pin at some intermediate point along its length, Fig. 5-4c, then this provides a condition of zero moment, and so a determinate structural analysis of the girder can be performed.

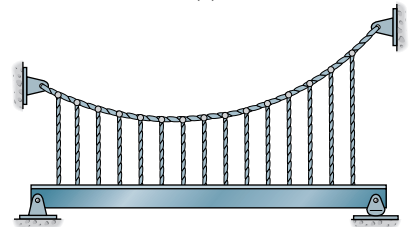


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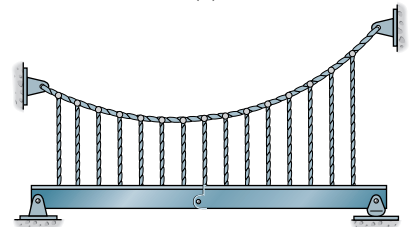
The Verrazano-Narrows Bridge at the entrance to New York Harbor has a main span of 4260 ft (1.30 km).



(a)



(b)



(c)

Fig. 5-4

EXAMPLE 5.2

The cable in Fig. 5-5a supports a girder which weighs 850 lb/ft. Determine the tension in the cable at points A, B, and C.

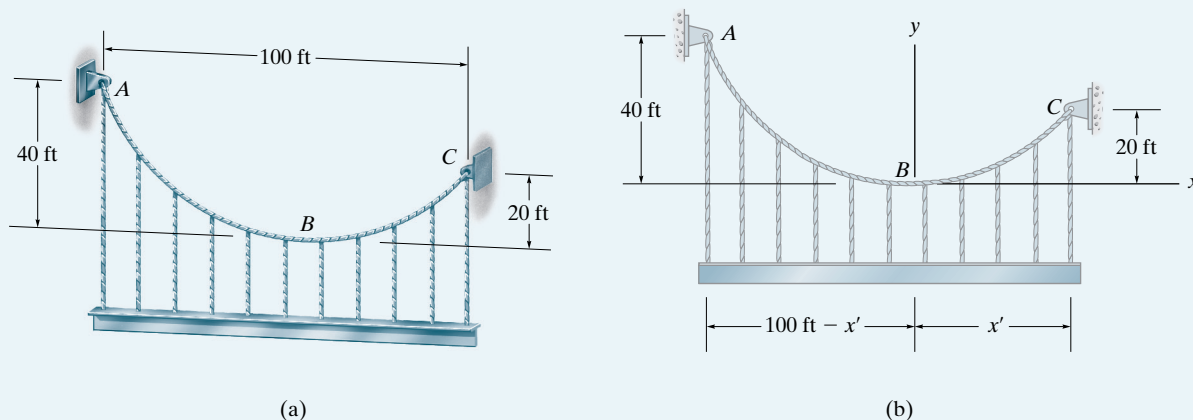


Fig. 5-5

SOLUTION

The origin of the coordinate axes is established at point B, the lowest point on the cable, where the slope is zero, Fig. 5-5b. From Eq. 5-7, the parabolic equation for the cable is:

$$y = \frac{w_0}{2F_H} x^2 = \frac{850 \text{ lb/ft}}{2F_H} x^2 = \frac{425}{F_H} x^2 \quad (1)$$

Assuming point C is located x' from B, we have

$$\begin{aligned} 20 &= \frac{425}{F_H} x'^2 \\ F_H &= 21.25x'^2 \end{aligned} \quad (2)$$

Also, for point A,

$$\begin{aligned} 40 &= \frac{425}{F_H} [-(100 - x')]^2 \\ 40 &= \frac{425}{21.25x'^2} [-(100 - x')]^2 \\ x'^2 + 200x' - 10\,000 &= 0 \\ x' &= 41.42 \text{ ft} \end{aligned}$$

Thus, from Eqs. 2 and 1 (or Eq. 5-6) we have

$$F_H = 21.25(41.42)^2 = 36\,459.2 \text{ lb}$$

$$\frac{dy}{dx} = \frac{850}{36\,459.2}x = 0.02331x \quad (3)$$

At point A,

$$x = -(100 - 41.42) = -58.58 \text{ ft}$$

$$\tan \theta_A = \left. \frac{dy}{dx} \right|_{x=-58.58} = 0.02331(-58.58) = -1.366$$

$$\theta_A = -53.79^\circ$$

Using Eq. 5-4,

$$T_A = \frac{F_H}{\cos \theta_A} = \frac{36\,459.2}{\cos(-53.79^\circ)} = 61.7 \text{ k} \quad \text{Ans.}$$

At point B, $x = 0$,

$$\tan \theta_B = \left. \frac{dy}{dx} \right|_{x=0} = 0, \quad \theta_B = 0^\circ$$

$$T_B = \frac{F_H}{\cos \theta_B} = \frac{36\,459.2}{\cos 0^\circ} = 36.5 \text{ k} \quad \text{Ans.}$$

At point C,

$$x = 41.42 \text{ ft}$$

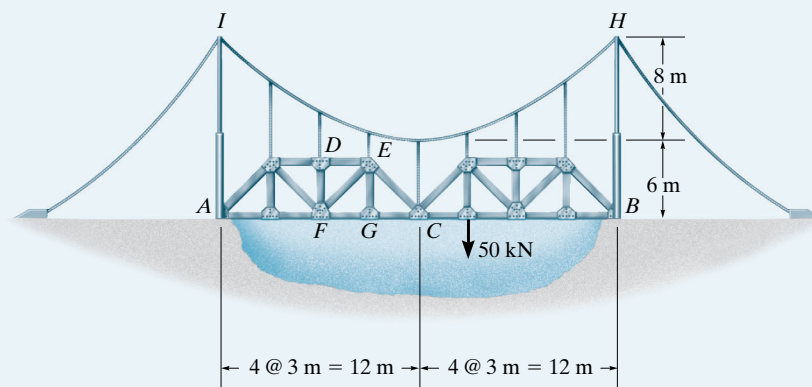
$$\tan \theta_C = \left. \frac{dy}{dx} \right|_{x=41.42} = 0.02331(41.42) = 0.9657$$

$$\theta_C = 44.0^\circ$$

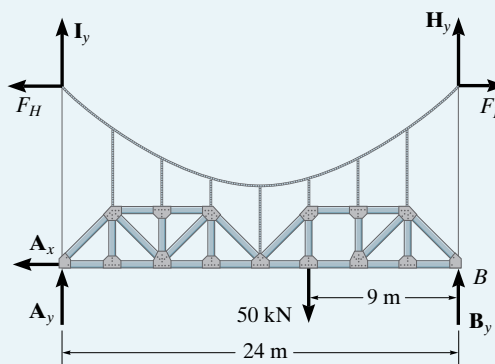
$$T_C = \frac{F_H}{\cos \theta_C} = \frac{36\,459.2}{\cos 44.0^\circ} = 50.7 \text{ k} \quad \text{Ans.}$$

EXAMPLE 5.3

The suspension bridge in Fig. 5-6a is constructed using the two stiffening trusses that are pin connected at their ends C and supported by a pin at A and a rocker at B . Determine the maximum tension in the cable IH . The cable has a parabolic shape and the bridge is subjected to the single load of 50 kN.



(a)



(b)

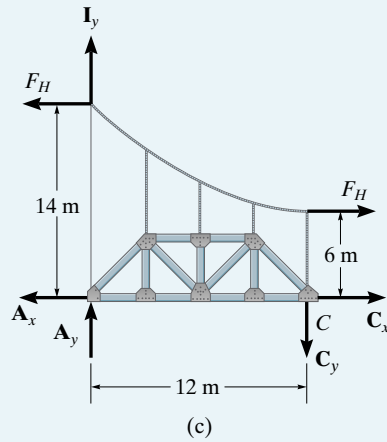
Fig. 5-6

SOLUTION

The free-body diagram of the cable-truss system is shown in Fig. 5-6b. According to Eq. 5-4 ($T \cos \theta = F_H$), the horizontal component of cable tension at I and H must be constant, F_H . Taking moments about B , we have

$$\zeta + \sum M_B = 0; \quad -I_y(24 \text{ m}) - A_y(24 \text{ m}) + 50 \text{ kN}(9 \text{ m}) = 0$$

$$I_y + A_y = 18.75$$



If only half the suspended structure is considered, Fig. 5-6c, then summing moments about the pin at C, we have

$$\zeta + \sum M_C = 0; \quad F_H(14 \text{ m}) - F_H(6 \text{ m}) - I_y(12 \text{ m}) - A_y(12 \text{ m}) = 0$$

$$I_y + A_y = 0.667F_H$$

From these two equations,

$$18.75 = 0.667F_H$$

$$F_H = 28.125 \text{ kN}$$

To obtain the maximum tension in the cable, we will use Eq. 5-11, but first it is necessary to determine the value of an assumed uniform distributed loading w_0 from Eq. 5-8:

$$w_0 = \frac{2F_H h}{L^2} = \frac{2(28.125 \text{ kN})(8 \text{ m})}{(12 \text{ m})^2} = 3.125 \text{ kN/m}$$

Thus, using Eq. 5-11, we have

$$\begin{aligned} T_{\max} &= w_0 L \sqrt{1 + (L/2h)^2} \\ &= 3.125(12 \text{ m}) \sqrt{1 + (12 \text{ m}/2(8 \text{ m}))^2} \\ &= 46.9 \text{ kN} \end{aligned}$$

Ans.